

KENAI, AK, USA

WMO: 702590

Lat: 60.580N

Lon: 151.239W

Elev: 91

StdP: 14.65

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26523

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.6	-13.9	-27.0	1.2	-21.5	-19.1	1.9	-13.6	27.8	28.0	25.8	26.5	2.1	10	0.744

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	70.5	57.7	67.2	56.4	64.5	55.4	59.8	67.2	58.3	64.5	56.9	62.4	9.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.6	68.5	61.4	55.2	65.0	60.0	54.1	62.5	59.0	26.2	67.2	25.2	64.6	24.3	62.3	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.3	21.7	19.0	DB	-23.9	76.9	7.8	4.3	-29.5	79.9	-34.0	82.4	-38.4	84.8	-44.0	87.9	(4)
(5)			WB	-24.0	62.4	7.5	3.5	-29.5	64.9	-33.9	67.0	-38.1	68.9	-43.6	71.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.3	15.9	20.8	24.4	36.5	45.8	52.5	56.3	55.1	48.4	36.6	23.3	18.4
	DBStd	17.12	15.17	11.85	9.98	6.55	4.71	3.72	3.08	3.75	5.00	8.58	11.32	13.03
	HDD50	5520	1059	818	794	404	145	14	0	5	84	418	801	978
	HDD65	10494	1524	1238	1259	854	595	374	270	307	498	881	1251	1443
	CDD50	502	0	0	0	0	15	90	196	163	36	2	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	33	0	0	0	0	2	10	13	8	0	0	0	0
	CDH80	4	0	0	0	0	0	1	2	1	0	0	0	0
Wind	WSAvg	7.9	7.6	8.1	8.6	8.2	8.6	8.4	7.8	7.0	7.1	7.3	7.5	8.1
Precipitation	PrecAvg	19.0	1.1	1.0	0.9	0.7	1.0	1.1	1.7	2.6	3.1	2.6	1.6	1.5
	PrecMax	27.1	3.0	2.8	3.2	1.8	2.8	2.4	3.6	5.4	7.1	7.4	6.9	4.0
	PrecMin	11.0	0.1	0.2	0.0	0.0	0.1	0.1	0.3	0.6	0.7	0.7	0.0	0.2
	PrecStd	3.9	0.8	0.6	0.8	0.6	0.7	0.6	0.9	1.3	1.6	1.4	1.3	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.5	42.2	46.4	58.4	71.8	74.5	75.0	74.4	66.1	57.2	46.0	41.7
		MCWB	37.5	37.6	39.0	45.7	53.7	58.1	60.5	61.0	54.7	49.9	41.2	38.0
	2%	DB	38.6	39.1	42.3	52.9	64.5	68.5	69.4	69.4	62.2	51.9	42.5	38.7
		MCWB	35.6	35.6	36.7	42.9	50.4	55.6	58.4	58.6	52.9	46.6	39.1	35.8
	5%	DB	36.4	36.9	39.2	50.1	59.6	64.3	66.3	66.2	59.5	49.6	39.8	36.5
		MCWB	34.0	34.1	34.4	41.5	47.7	53.5	57.4	56.7	51.9	44.7	37.5	34.2
	10%	DB	34.3	34.9	37.0	47.2	56.0	61.1	63.9	63.5	57.1	47.6	37.5	34.4
		MCWB	32.4	32.7	33.3	40.1	46.1	52.1	56.5	55.9	51.2	43.6	35.3	32.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.2	38.3	40.1	47.7	55.8	59.6	62.2	63.0	56.5	51.1	42.1	38.6
		MCDB	41.1	41.1	45.0	55.1	69.1	71.6	71.2	72.0	62.7	55.1	44.8	41.0
	2%	WB	36.1	36.1	37.3	44.2	51.2	56.7	59.8	60.1	54.5	47.8	39.8	36.3
		MCDB	38.2	38.6	41.2	51.3	61.8	66.0	67.2	67.1	59.5	50.8	41.8	38.1
	5%	WB	34.3	34.4	35.4	42.1	49.0	54.6	58.4	58.4	53.3	46.0	37.7	34.5
		MCDB	36.1	36.4	38.5	48.5	57.6	62.1	64.6	64.0	57.4	48.8	39.6	36.1
	10%	WB	32.4	32.9	33.8	40.6	47.2	53.1	57.2	56.9	52.0	44.2	35.6	32.8
		MCDB	34.2	34.7	36.5	46.4	54.5	59.5	62.5	61.6	55.9	47.0	37.4	34.3
Mean Daily Temperature Range	5% DB	MDBR	13.8	15.1	17.1	15.9	17.2	15.2	13.4	15.1	15.0	13.7	13.4	13.4
		MCDBR	9.7	11.2	15.1	19.9	25.9	23.8	20.3	22.3	19.4	13.9	9.6	9.7
		MCWBR	8.3	9.0	10.9	12.0	14.1	12.7	11.2	12.8	12.2	9.6	7.7	8.3
	5% WB	MCDBR	9.6	10.2	12.9	17.7	21.9	19.9	16.9	18.0	15.3	11.7	9.2	9.8
		MCWBR	8.5	8.4	9.6	11.5	12.7	11.1	10.2	11.1	10.3	8.6	7.5	8.7
Clear-Sky Solar Irradiance	taub	0.239	0.285	0.303	0.343	0.367	0.358	0.366	0.351	0.323	0.300	0.251	0.225	
	taud	2.168	2.296	2.330	2.330	2.340	2.451	2.460	2.504	2.562	2.494	2.174	1.982	
	Ebn at Noon	168	227	263	271	272	277	271	267	257	224	157	109	
	Edh at Noon	12	20	27	33	35	32	31	28	22	17	12	9	
All-Sky Solar Radiation	RadAvg	94	306	797	1286	1647	1722	1516	1205	781	372	134	47	
	RadStd	12	43	93	127	136	124	149	144	111	39	22	6	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.48	+6.32	+6.73	N/A	N/A	N/A	-488	-578	N/A	N/A
(47) Regional (22 neighbors)		+1.22	+3.87	+3.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page